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## JEE MAIN PHYSICS DPP: KINEMATICS 1D

Topic: Frame of Reference, Position, Distance, Displacement & 1D Motion | Time: 45 Min | Max Marks: 80

Marking Scheme: +4, -1 | Designed & Curated by: Syed Mazhar Ali

- Q1.** A particle moves along a circular path of radius  $R$  from point  $A$  to point  $B$  subtending an angle of  $120^\circ$  at the centre. The ratio of distance travelled to the magnitude of displacement is:
- $\frac{2\pi}{3\sqrt{3}}$
  - $\frac{\pi}{\sqrt{3}}$
  - $\frac{2\pi}{3}$
  - $\frac{\pi}{3\sqrt{3}}$
- Q2.** A body starts moving from the origin along the  $x$ -axis according to  $x(t) = 6t^2 - t^3$ , where  $x$  is in metres and  $t$  is in seconds. The total distance travelled by the body before it momentarily comes to rest is:
- 32 m
  - 64 m
  - 16 m
  - 48 m
- Q3.** A person travelling along a straight road moves with speed  $v_1$  for the first one-third of the total distance and with speed  $v_2$  for the remaining two-thirds of the distance. The average speed of the person is:
- $\frac{3v_1v_2}{2v_1 + v_2}$
  - $\frac{3v_1v_2}{v_1 + 2v_2}$
  - $\frac{v_1 + 2v_2}{3}$
  - $\frac{2v_1v_2}{v_1 + v_2}$
- Q4.** A car travels along a straight line. For the first half of the total time of journey it moves with speed 40 km/h and for the second half time with speed 60 km/h. The average speed of the car is:
- 48 km/h
  - 50 km/h
  - 52 km/h
  - 45 km/h
- Q5.** The position  $x$  of a particle moving along a straight line varies with time  $t$  as  $x = \sqrt{at^2 + 2bt + c}$ . The acceleration of the particle is proportional to:
- $x^{-3}$
  - $x^{-2}$
  - $x^3$
  - $x^{-1}$
- Q6.** Which of the following statements is **always true** for one-dimensional rectilinear motion?
- Magnitude of displacement is equal to the total path length.
  - If speed is constant, acceleration must be zero.
  - If velocity is zero at an instant, acceleration must also be zero at that instant.
  - Average velocity is equal to the average of initial and final velocities.
- Q7.** A particle moves along the  $x$ -axis such that its position is given by  $x(t) = t^3 - 6t^2 + 9t + 2$  m. The total distance covered by the particle from  $t = 0$  to  $t = 5$  s is:
- 20 m
  - 28 m
  - 24 m
  - 32 m
- Q8.** A passenger in a train moving at a constant velocity drops a coin. To the passenger, the trajectory of the coin appears as a vertical straight line. To an observer standing on the ground platform, the trajectory of the coin appears as a:
- Vertical straight line
  - Parabola
  - Circle
  - Hyperbola
- Q9.** The displacement  $x$  of a particle moving in one dimension under the action of a constant force is related to time  $t$  by  $t = \sqrt{x} + 3$ , where  $x$  is in metres and  $t$  is in seconds. The displacement of the particle when its velocity is zero is:
- 0 m
  - 3 m
  - 9 m
  - 6 m
- Q10.** A particle moves in a straight line with constant deceleration. It covers 10 m in the 1st second and 6 m in the 2nd second. The particle comes to rest at time  $t$  equal to:
- 3.0 s
  - 3.5 s
  - 4.0 s
  - 4.5 s

**Q11.** A ball is thrown vertically upwards from the ground with speed  $u$ . It passes a point at height  $h$  at time  $t_1$  on its way up and at time  $t_2$  on its way down. The height  $h$  is given by:

- (a)  $\frac{1}{2}gt_1t_2$
- (b)  $gt_1t_2$
- (c)  $\frac{1}{4}gt_1t_2$
- (d)  $2gt_1t_2$

**Q12.** A point moves along a straight line such that its velocity is  $v = v_0(1 - t/T)$  for  $0 \leq t \leq T$ . The total distance travelled by the particle in time  $T$  is:

- (a)  $\frac{1}{2}v_0T$
- (b)  $v_0T$
- (c)  $\frac{1}{3}v_0T$
- (d)  $\frac{1}{4}v_0T$

**Q13.** An elevator is accelerating upwards with an acceleration  $a = 2 \text{ m/s}^2$ . A bolt falls from the ceiling of the elevator from a height of 2.4 m above the floor. Taking  $g = 10 \text{ m/s}^2$ , the time taken by the bolt to reach the floor is:

- (a)  $\sqrt{0.4} \text{ s} \approx 0.63 \text{ s}$
- (b)  $0.4 \text{ s}$
- (c)  $\sqrt{0.6} \text{ s} \approx 0.77 \text{ s}$
- (d)  $0.8 \text{ s}$

**Q14.** The acceleration  $a$  of a particle moving along a straight line is given by  $a = -kv^2$ , where  $k$  is a positive constant and  $v$  is velocity. If initial velocity is  $v_0$  at  $t = 0$ , the velocity at time  $t$  is:

- (a)  $\frac{v_0}{1 + kv_0t}$
- (b)  $v_0e^{-kt}$
- (c)  $\frac{v_0}{1 - kv_0t}$
- (d)  $v_0(1 - kv_0t)$

**Q15.** A particle moves along a straight line such that its velocity depends on displacement  $s$  as  $v = \alpha\sqrt{s}$ , where  $\alpha$  is a positive constant. The acceleration of the particle is:

- (a)  $\frac{\alpha^2}{2}$
- (b)  $\alpha^2$
- (c)  $\frac{\alpha}{2}$
- (d)  $\frac{\alpha^2}{\sqrt{s}}$

**Q16.** A body dropped from a tower of height  $H$  reaches the ground in  $T$  seconds. The position of the body at time  $t = T/3$  from the top of the tower is:

- (a)  $\frac{H}{9}$  below the top
- (b)  $\frac{8H}{9}$  below the top
- (c)  $\frac{H}{3}$  below the top
- (d)  $\frac{2H}{3}$  below the top

**Q17.** For an object moving along a straight line, the ratio of magnitude of average velocity to average speed over any given time interval is:

- (a) Always = 1
- (b) Always  $\geq 1$
- (c) Always  $\leq 1$
- (d) Can be greater or less than 1

**Q18.** A train of length 200 m travelling at 72 km/h overtakes another train of length 300 m travelling at 36 km/h in the same direction on parallel tracks. The time taken to completely cross is:

- (a) 50 s
- (b) 25 s
- (c) 100 s
- (d) 75 s

**Q19.** The velocity-time relation of a body is  $v = 3t^2 - 12t + 9 \text{ m/s}$ . The body changes its direction of motion at times:

- (a)  $t = 1 \text{ s}$  and  $t = 3 \text{ s}$
- (b)  $t = 2 \text{ s}$  only
- (c)  $t = 3 \text{ s}$  only
- (d)  $t = 0 \text{ s}$  and  $t = 4 \text{ s}$

**Q20.** A bullet loses  $\frac{1}{20}$ th of its velocity in passing through a wooden plank of thickness  $d$ . The minimum number of identical planks required to stop the bullet completely is:

- (a) 11
- (b) 10
- (c) 20
- (d) 19

# ANSWER KEY & STEP-BY-STEP SOLUTIONS

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## Quick Answer Key

| Q.No | Ans | Q.No | Ans | Q.No | Ans | Q.No | Ans |
|------|-----|------|-----|------|-----|------|-----|
| 1    | (a) | 6    | (b) | 11   | (a) | 16   | (a) |
| 2    | (a) | 7    | (b) | 12   | (a) | 17   | (c) |
| 3    | (a) | 8    | (b) | 13   | (a) | 18   | (a) |
| 4    | (d) | 9    | (a) | 14   | (a) | 19   | (a) |
| 5    | (a) | 10   | (a) | 15   | (a) | 20   | (a) |

## Detailed Solutions

### Sol 1. Correct Option: (a)

- Distance along arc:  $s = R\theta = R\left(\frac{2\pi}{3}\right) = \frac{2\pi R}{3}$ .
- Displacement magnitude:

$$|\Delta \vec{r}| = 2R \sin\left(\frac{\theta}{2}\right) = 2R \sin(60^\circ) = \sqrt{3}R$$

Ratio:

$$\frac{\text{Distance}}{\text{Displacement}} = \frac{2\pi R/3}{\sqrt{3}R} = \frac{2\pi}{3\sqrt{3}}$$

### Sol 2. Correct Option: (a)

- $x(t) = 6t^2 - t^3 \implies v(t) = \frac{dx}{dt} = 12t - 3t^2$ .
- Setting  $v(t) = 0 \implies 3t(4 - t) = 0 \implies t = 4$  s.
- Since velocity is positive for all  $0 < t < 4$  s, motion is unidirectional.
- Distance =  $x(4) = 6(4)^2 - (4)^3 = 96 - 64 = 32$  m.

### Sol 3. Correct Option: (a)

- Let total distance be  $s$ .
- $t_1 = \frac{s/3}{v_1} = \frac{s}{3v_1}$ ,  $t_2 = \frac{2s/3}{v_2} = \frac{2s}{3v_2}$ .
- Average speed:

$$v_{\text{avg}} = \frac{s}{t_1 + t_2} = \frac{s}{\frac{s}{3} \left( \frac{1}{v_1} + \frac{2}{v_2} \right)}$$

$$= \frac{3}{\frac{v_2 + 2v_1}{v_1 v_2}} = \frac{3v_1 v_2}{2v_1 + v_2}$$

### Sol 4. Correct Option: (b)

- Let total time be  $2t_0$ .
- Distance  $s = s_1 + s_2 = (40)(t_0) + (60)(t_0) = 100t_0$ .
- Average speed:

$$v_{\text{avg}} = \frac{s}{2t_0} = \frac{100t_0}{2t_0} = 50 \text{ km/h}$$

### Sol 5. Correct Option: (a)

- $x^2 = at^2 + 2bt + c$ .
- Differentiating with respect to  $t$ :
- $$2x \frac{dx}{dt} = 2at + 2b \implies xv = at + b$$

Differentiating again:

$$x \frac{dv}{dt} + v \left( \frac{dx}{dt} \right) = a \implies xa_{\text{acc}} + v^2 = a$$

$$xa_{\text{acc}} = a - v^2 = a - \left( \frac{at + b}{x} \right)^2 = \frac{ax^2 - (at + b)^2}{x^2}$$

Since  $ax^2 - (at + b)^2 = a(at^2 + 2bt + c) - (a^2t^2 + 2abt + b^2) = ac - b^2$  (constant):

$$a_{\text{acc}} = \frac{ac - b^2}{x^3} \propto x^{-3}$$

### Sol 6. Correct Option: (b)

In one-dimensional motion along a straight line, speed is the magnitude of velocity  $|v|$ . If speed is constant and direction cannot change without passing through zero speed, velocity vector is constant, hence acceleration is strictly zero.

### Sol 7. Correct Option: (b)

- $v(t) = \frac{dx}{dt} = 3t^2 - 12t + 9 = 3(t - 1)(t - 3)$ .
- The particle reverses direction at  $t = 1$  s and  $t = 3$  s.
- $x(0) = 2$  m
  - $x(1) = 1 - 6 + 9 + 2 = 6$  m
  - $x(3) = 27 - 54 + 27 + 2 = 2$  m
  - $x(5) = 125 - 150 + 45 + 2 = 22$  m.
- Total Distance:

$$|x(1) - x(0)| + |x(3) - x(1)| + |x(5) - x(3)|$$

$$= |6 - 2| + |2 - 6| + |22 - 2| = 4 + 4 + 20 = 28 \text{ m}$$

### Sol 8. Correct Option: (b)

In the platform frame, the coin possesses initial horizontal velocity equal to the train's speed  $v$  and vertical downward acceleration  $g$ . The combination of horizontal constant velocity ( $x = vt$ ) and vertical accelerated motion ( $y = \frac{1}{2}gt^2$ ) traces a parabola ( $y = \frac{g}{2v^2}x^2$ ).

### Sol 9. Correct Option: (a)

- $\sqrt{x} = t - 3 \implies x = (t - 3)^2$ .
- $v(t) = \frac{dx}{dt} = 2(t - 3)$ .
- Velocity is zero at  $t = 3$  s.
- Displacement at  $t = 3$  s is  $x(3) = (3 - 3)^2 = 0$  m.

**Sol 10. Correct Option: (a)**

$$s_n = u - \frac{a}{2}(2n - 1).$$

$$s_1 = u - \frac{a}{2}(1) = 10 \implies u - 0.5a = 10.$$

$$s_2 = u - \frac{a}{2}(3) = 6 \implies u - 1.5a = 6.$$

$$\text{Subtracting: } a = 4 \text{ m/s}^2 \implies u = 12 \text{ m/s}.$$

$$\text{Time to stop: } t = \frac{u}{a} = \frac{12}{4} = 3.0 \text{ s}.$$

**Sol 11. Correct Option: (a)**

$$\text{Equation of motion: } h = ut - \frac{1}{2}gt^2 \implies \frac{1}{2}gt^2 - ut + h = 0.$$

Since  $t_1$  and  $t_2$  are the roots of this quadratic equation, product of roots gives:

$$t_1 t_2 = \frac{h}{\frac{1}{2}g} \implies h = \frac{1}{2}gt_1 t_2$$

**Sol 12. Correct Option: (a)**

For  $0 \leq t \leq T$ , velocity  $v(t) \geq 0$ , so distance equals displacement:

$$s = \int_0^T v_0 \left(1 - \frac{t}{T}\right) dt = v_0 \left[t - \frac{t^2}{2T}\right]_0^T = \frac{1}{2}v_0 T$$

**Sol 13. Correct Option: (a)**

In the elevator reference frame (pseudo-force downwards):

$$g_{\text{eff}} = g + a = 10 + 2 = 12 \text{ m/s}^2$$

$$\text{Using } h = \frac{1}{2}g_{\text{eff}}t^2:$$

$$2.4 = \frac{1}{2}(12)t^2 = 6t^2$$

$$t^2 = \frac{2.4}{6} = 0.4 \implies t = \sqrt{0.4} \text{ s}$$

**Sol 14. Correct Option: (a)**

$$\frac{dv}{dt} = -kv^2 \implies -\frac{dv}{v^2} = kdt.$$

Integrating from  $v_0$  to  $v$ :

$$\left[\frac{1}{v}\right]_{v_0}^v = kt \implies \frac{1}{v} - \frac{1}{v_0} = kt \implies v = \frac{v_0}{1 + kv_0 t}$$

**Sol 15. Correct Option: (a)**

$$v = \alpha s^{1/2} \implies \frac{dv}{ds} = \frac{1}{2}\alpha s^{-1/2}.$$

Acceleration:

$$a = v \frac{dv}{ds} = (\alpha s^{1/2}) \left(\frac{1}{2}\alpha s^{-1/2}\right) = \frac{\alpha^2}{2}$$

**Sol 16. Correct Option: (a)**

$$H = \frac{1}{2}gT^2.$$

Distance fallen in time  $t = T/3$ :

$$y = \frac{1}{2}g \left(\frac{T}{3}\right)^2 = \frac{1}{9} \left(\frac{1}{2}gT^2\right) = \frac{H}{9}$$

Thus, it is at a distance  $H/9$  below the top.

**Sol 17. Correct Option: (c)**

Since distance  $\geq$  |displacement| for any path, average speed  $\geq$  |average velocity|.

$$\text{Thus, } \frac{|\text{Average Velocity}|}{\text{Average Speed}} \leq 1.$$

**Sol 18. Correct Option: (a)**

$$\text{Relative speed: } v_{\text{rel}} = 72 - 36 = 36 \text{ km/h} =$$

$$36 \times \frac{5}{18} = 10 \text{ m/s}.$$

$$\text{Total distance to clear: } L = 200 + 300 = 500 \text{ m}.$$

$$\text{Time: } t = \frac{L}{v_{\text{rel}}} = \frac{500}{10} = 50 \text{ s}.$$

**Sol 19. Correct Option: (a)**

$$v(t) = 3(t^2 - 4t + 3) = 3(t - 1)(t - 3).$$

$v(t)$  crosses zero and changes sign at  $t = 1$  s and  $t = 3$  s.

**Sol 20. Correct Option: (a)**

Using  $v^2 = u^2 - 2as$ :

$$v_1 = u - \frac{u}{20} = \frac{19}{20}u.$$

$$\left(\frac{19}{20}u\right)^2 = u^2 - 2ad \implies 2ad = u^2 \left(1 - \frac{361}{400}\right) = \frac{39}{400}u^2.$$

Let  $n$  planks stop the bullet ( $v = 0$  over distance  $nd$ ):

$$0 = u^2 - 2a(nd) \implies n(2ad) = u^2$$

$$n \left(\frac{39}{400}u^2\right) = u^2 \implies n = \frac{400}{39} \approx 10.25$$

Hence, a minimum of **11** planks are required.

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